ResearchX

Intelligent Research Discovery & Project Similarity Platform

DDL commands

ResearchX — PostgreSQL DDL

```sql
-- ============================================================
-- ResearchX - Relational Schema DDL (PostgreSQL)
-- Scope: relational core only (PostgreSQL layer of the architecture)
-- ============================================================

CREATE TABLE role (
    role_id       SMALLINT PRIMARY KEY,
    role_name     VARCHAR(30) NOT NULL UNIQUE,
    CHECK (role_name IN ('Admin','Student','Faculty'))
);

CREATE TABLE app_user (
    user_id       SERIAL PRIMARY KEY,
    username      VARCHAR(50) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    full_name     VARCHAR(100) NOT NULL,
    email         VARCHAR(120) NOT NULL UNIQUE,
    role_id       SMALLINT NOT NULL REFERENCES role(role_id),
    created_at    TIMESTAMP NOT NULL DEFAULT now()
);

CREATE TABLE domain (
    domain_id     SERIAL PRIMARY KEY,
    domain_name   VARCHAR(150) NOT NULL UNIQUE,
    description   TEXT
);

CREATE TABLE project (
    project_id       SERIAL PRIMARY KEY,
    student_id       INTEGER NOT NULL REFERENCES app_user(user_id),
    title            VARCHAR(300) NOT NULL,
    problem_statement TEXT NOT NULL,
    description      TEXT,
    status           VARCHAR(30) NOT NULL DEFAULT 'Pending',
    approved_by      INTEGER REFERENCES app_user(user_id),
    created_at       TIMESTAMP NOT NULL DEFAULT now(),
    CHECK (status IN ('Pending','Approved','Rejected'))
);

CREATE TABLE project_review (
    review_id     SERIAL PRIMARY KEY,
    project_id    INTEGER NOT NULL REFERENCES project(project_id)
                  ON DELETE CASCADE,
    faculty_id    INTEGER NOT NULL REFERENCES app_user(user_id),
    decision      VARCHAR(30) NOT NULL,
    comments      TEXT,
    review_date   TIMESTAMP NOT NULL DEFAULT now(),
    CHECK (decision IN ('Approved','Rejected','Changes Required'))
);

CREATE TABLE author (
    author_id     SERIAL PRIMARY KEY,
    name          VARCHAR(150) NOT NULL,
    affiliation   VARCHAR(200),
    email         VARCHAR(120) UNIQUE
);

CREATE TABLE research_paper (
    paper_id       SERIAL PRIMARY KEY,
    title          VARCHAR(500) NOT NULL,
    abstract       TEXT,
    keywords       TEXT,
    doi            VARCHAR(255) UNIQUE,
    file_url       VARCHAR(500),
    published_in   VARCHAR(255),
    year           SMALLINT,
    cited_by_count INTEGER NOT NULL DEFAULT 0
                   CHECK (cited_by_count >= 0)
);

CREATE TABLE paper_author (
    paper_id      INTEGER NOT NULL REFERENCES research_paper(paper_id)
                  ON DELETE CASCADE,
    author_id     INTEGER NOT NULL REFERENCES author(author_id)
                  ON DELETE CASCADE,
    author_order  SMALLINT,
    PRIMARY KEY (paper_id, author_id)
);
```

ResearchX — DDL commands (continued)

ResearchX — PostgreSQL DDL

```sql
CREATE TABLE citation (
    citation_id      SERIAL PRIMARY KEY,
    citing_paper_id  INTEGER NOT NULL REFERENCES research_paper(paper_id)
                     ON DELETE CASCADE,
    cited_paper_id   INTEGER NOT NULL REFERENCES research_paper(paper_id)
                     ON DELETE CASCADE,
    UNIQUE (citing_paper_id, cited_paper_id),
    CHECK (citing_paper_id <> cited_paper_id)
);

CREATE TABLE saved_item (
    save_id     SERIAL PRIMARY KEY,
    user_id     INTEGER NOT NULL REFERENCES app_user(user_id)
                ON DELETE CASCADE,
    paper_id    INTEGER NOT NULL REFERENCES research_paper(paper_id)
                ON DELETE CASCADE,
    saved_at    TIMESTAMP NOT NULL DEFAULT now(),
    UNIQUE (user_id, paper_id)
);

-- ResearchX semantic-search support
-- Embedding vectors are stored in the vector database;
-- this table keeps the relational reference/metadata.
CREATE TABLE embedding_ref (
    embedding_id SERIAL PRIMARY KEY,
    entity_type  VARCHAR(30) NOT NULL,
    entity_id    INTEGER NOT NULL,
    vector_ref   VARCHAR(150) NOT NULL UNIQUE,
    model        VARCHAR(150) NOT NULL,
    created_at   TIMESTAMP NOT NULL DEFAULT now(),
    CHECK (entity_type IN ('PROJECT','RESEARCH_PAPER'))
);

-- Useful relational indexes
CREATE INDEX idx_project_student
    ON project(student_id);

CREATE INDEX idx_project_status
    ON project(status);

CREATE INDEX idx_paper_year
    ON research_paper(year);

CREATE INDEX idx_citation_citing
    ON citation(citing_paper_id);

CREATE INDEX idx_citation_cited
    ON citation(cited_paper_id);

-- Seed the role lookup table
INSERT INTO role (role_id, role_name) VALUES
    (1, 'Admin'),
    (2, 'Student'),
    (3, 'Faculty');
```